Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: M.Tech(CSE) 1st sem
Name of the Paper: Data Structures and Algorithms
Paper Code: MCS-141

Time: 1.5 hour | Maximum Marks: 50

Note:
- *(i)* Answer all the questions by choosing any one of the sub questions
- *(ii)* Each question carries 10 marks.
- *(iii)* Please specify COs against each question.

Q1. (10 Marks) (CO-1)

a. Determine the Big-O notation for an algorithm that performs nested loops where the outer loop runs n times and the inner loop runs n^2 times.

OR

b. Given the insertion sort algorithm, determine its best, worst, and average case complexities.

Q2. (10 Marks) (CO-1)

a. Use the Master Theorem to solve the recurrence T(n)=3T(n/4)+nT(n) .

OR

b. Under what circumstances would you prefer linear search over binary search? Justify your answer.

Q3. (10 Marks) (CO-1)

a. Write a recursive function to solve the Towers of Hanoi problem and compute its time complexity.

OR

b. How does the process of heapifying an array differ from inserting an element into a Max Heap?

Q4. (10 Marks) (CO-2)

a. Why is a linked list-based implementation preferred for a queue when dynamic resizing is needed?

OR

b. Compare the space utilization of a circular queue to a regular queue. How does the circular queue prevent space wastage in a static array-based implementation?

Q5. (10 Marks) (CO-2)

a. Given an infix expression with operators of varying precedence, analyze how the stack handles operator precedence and parentheses during conversion to postfix.

OR

b. When would you prefer using an adjacency matrix over an adjacency list and vice versa?